Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

November 8, 2017

Day 4: Random Variables: Continuous

- CDF
- Continuous Random Variables
- More on Expectation and Variance
- Brief introduction to inference

Pop Quiz

- If there are 5 people in the class and each has a 80 percent chance of getting a question right, what is the best guess for how many students will get the question right?
- What is the probability that exactly 2 people get the question right?

Binomial

$$\binom{n}{k} p^k (1-p)^{n-k}$$

- $n = 5,\ p = .8,\ k = 2$

$$\binom{5}{2} .8^2 (.2)^3$$

- $10 * .64 * .008 = .0512$
- dbinom(2,5,.8)

Binomial

$$\binom{n}{k} p^k (1-p)^{n-k}$$

- $n = 5,\ p = .8,\ k = 2$
$$\binom{5}{2} .8^2 (.2)^3$$
- $10 * .64 * .008 = .0512$
- dbinom(2,5,.8)

Cumulative Distribution Function

Definition

The Cumulative Distribution Function (CDF) of a random variable X is:

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$$

The CDF characterizes how probability cumulates as X gets larger.

Cumulative Distribution Function and the Probability Function

Definition

The Cumulative Distribution Function (CDF) of a random variable X is:

$$F_X(x) = \sum_{k \leq x} p_X(k)$$

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^{4} p_X(x)$$

$$F_X(4) = \sum_{k=0}^{4} \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is $F_X(22)$?
What is $F_X(-1)$?

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^{4} p_X(x)$$

$$F_X(4) = \sum_{k=0}^{4} \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is $F_X(22)$?
What is $F_X(-1)$?

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^{4} p_X(x)$$

$$F_X(4) = \sum_{k=0}^{4} \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is $F_X(22)$?
What is $F_X(-1)$?

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^{4} p_X(x)$$

$$F_X(4) = \sum_{k=0}^{4} \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is $F_X(22)$?
What is $F_X(-1)$?

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^{4} p_X(x)$$

$$F_X(4) = \sum_{k=0}^{4} \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is $F_X(22)$?

What is $F_X(-1)$?

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^{4} p_X(x)$$

$$F_X(4) = \sum_{k=0}^{4} \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is $F_X(22)$?
What is $F_X(-1)$?

Cumulative Distribution Function

Theorem

F(x) is a CDF if and only if:

1. $F_X(x)$ *is nondecreasing as x increases:*

$$x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$$

2. $F_X(x)$ *has limits at* $\pm\infty$

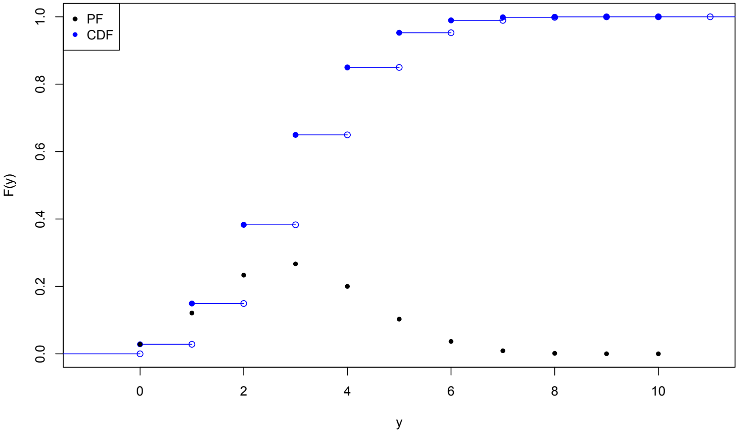
$$\lim_{x \to -\infty} F_X(x) = 0$$

$$\lim_{x \to \infty} F_X(x) = 1$$

3. *Continuity from the right:*

$$F_X(x) = \lim_{y \to x,\ y > x} F(x) \quad \forall x$$

CDF



Functions of Random Variables

We might (or often) apply a function to a random variable $g(X)$.

Functions of Random Variables

We might (or often) apply a function to a random variable $g(X)$.
How do we compute $E[g(X)]$?

Functions of Random Variables

We might (or often) apply a function to a random variable $g(X)$.
How do we compute $E[g(X)]$?

Proposition

Functions of Random Variables

We might (or often) apply a function to a random variable $g(X)$.
How do we compute $E[g(X)]$?

Proposition

Expected value of a function of a random variable: Suppose X is a discrete random variable that takes on values x_i, $i = \{1, 2, \ldots, \}$, with probabilities $p(x_i)$.

Functions of Random Variables

We might (or often) apply a function to a random variable $g(X)$. How do we compute $E[g(X)]$?

Proposition

Expected value of a function of a random variable: Suppose X is a discrete random variable that takes on values x_i, $i = \{1, 2, \ldots, \}$, with probabilities $p(x_i)$. If $g : X \to \mathcal{R}$, then its expected value $E[g(X)]$ is,

Functions of Random Variables

We might (or often) apply a function to a random variable $g(X)$.
How do we compute $E[g(X)]$?

Proposition

Expected value of a function of a random variable: Suppose X is a discrete random variable that takes on values x_i, $i = \{1, 2, \ldots, \}$, with probabilities $p(x_i)$. If $g : X \rightarrow \mathcal{R}$, then its expected value $E[g(X)]$ is,

$$E[g(X)] \;=\; \sum_i g(x_i)p(x_i)$$

Functions of Random Variables: Corollary

Corollary

Suppose X is a random variable and a and b are constants (not random variables). Then,

$$E[aX + b] = aE[X] + b$$

Proof.

Functions of Random Variables: Corollary

Corollary

Suppose X is a random variable and a and b are constants (not random variables). Then,

$$E[aX + b] = aE[X] + b$$

Proof.

$$E[aX + b] = \sum_{x:p(x)>0} (ax + b)p(x)$$

Functions of Random Variables: Corollary

Corollary

Suppose X is a random variable and a and b are constants (not random variables). Then,

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= \sum_{x:p(x)>0} axp(x) + \sum_{x:p(x)>0} bp(x) \end{aligned}$$

Functions of Random Variables: Corollary

Corollary

Suppose X is a random variable and a and b are constants (not random variables). Then,

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned}
E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\
&= \sum_{x:p(x)>0} axp(x) + \sum_{x:p(x)>0} bp(x) \\
&= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x)
\end{aligned}$$

Functions of Random Variables: Corollary

Corollary

Suppose X is a random variable and a and b are constants (not random variables). Then,

$$E[aX + b] \;=\; aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= \sum_{x:p(x)>0} axp(x) + \sum_{x:p(x)>0} bp(x) \\ &= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x) \\ &= aE[X] + b(1) \end{aligned}$$

Inequalities

Definition

Markov's Inequality Let X be a random variable, and g be a nonnegative function, then for any positive a:

$$P(g(X) \geq a) \leq \frac{E[g(X)]}{a}$$

Definition

Jensen's Inequality Let X be a random variable with $E(|X|) < \infty$. If g is convex, then

$$E[g(X)] \geq g(E(X))$$

Example of Chebychev

If X is a random variable, with mean μ and variance σ^2:

$$P(|X - \mu| \geq 2\sigma) \leq .25$$

$$P(|X - \mu| \geq 2\sigma) = P(|X - \mu|^2 \geq 4\sigma^2)$$

$$g(X) = |X - \mu|^2$$

$$E(g(X)) = \sum |X - \mu|^2 p_x = Var(X) = \sigma^2$$

$$P(|X - \mu|^2 \geq 4\sigma^2) \leq \frac{\sigma^2}{4\sigma^2}$$

$$P(|X - \mu| \geq 2\sigma) \leq \frac{1}{4}$$

Example of Chebychev

If X is a random variable, with mean μ and variance σ^2:

$$P(|X - \mu| \geq 2\sigma) \leq .25$$

$$P(|X - \mu| \geq 2\sigma) = P(|X - \mu|^2 \geq 4\sigma^2)$$
$$g(X) = |X - \mu|^2$$

$$E(g(X)) = \sum |X - \mu|^2 p_x = Var(X) = \sigma^2$$

$$P(|X - \mu|^2 \geq 4\sigma^2) \leq \frac{\sigma^2}{4\sigma^2}$$

$$P(|X - \mu| \geq 2\sigma) \leq \frac{1}{4}$$

Example of Chebychev

If X is a random variable, with mean μ and variance σ^2:

$$P(|X-\mu| \geq 2\sigma) \leq .25$$

$$P(|X-\mu| \geq 2\sigma) = P(|X-\mu|^2 \geq 4\sigma^2)$$
$$g(X) = |X-\mu|^2$$

$$E(g(X)) = \sum |X-\mu|^2 p_x = Var(X) = \sigma^2$$
$$P(|X-\mu|^2 \geq 4\sigma^2) \leq \frac{\sigma^2}{4\sigma^2}$$
$$P(|X-\mu| \geq 2\sigma) \leq \frac{1}{4}$$

Famous Distributions

- Bernoulli
- Binomial
- Multinomial
- Poisson

Models of how world works.

Poisson Distribution

Definition

Suppose X is a random variable that takes on values $X \in \{0, 1, 2, \ldots, \}$ and that $P(X = k) = p(k)$ is,

$$p(k) \quad = \quad e^{-\lambda}\frac{\lambda^k}{k!}$$

for $k \in \{0, 1, \ldots, \}$ and 0 otherwise. Then we will say that X follows a Poisson distribution with rate parameter λ.

$$X \quad \sim \quad Poisson(\lambda)$$

Poisson Distribution

Properties:

1) It is a probability distribution.

Poisson Distribution

Properties:

1) It is a probability distribution.
   Recall the Taylor expansion of e^x

Poisson Distribution

Properties:

1) It is a probability distribution.
   Recall the Taylor expansion of e^x

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

Poisson Distribution

Properties:

1) It is a probability distribution.
   Recall the Taylor expansion of e^x

$$
\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\
e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} &= e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2!} + \dots)
\end{aligned}
$$

Poisson Distribution

Properties:

1) It is a probability distribution.
   Recall the Taylor expansion of e^x

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\ e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} &= e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2!} + \dots) \\ &= e^{-\lambda}(e^{\lambda}) = 1 \end{aligned}$$

Poisson Distribution

Properties:

Poisson Distribution

Properties:

2) $E[X] = \lambda$

Poisson Distribution

Properties:

2) $E[X] = \lambda$

$$E[X] = e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!}$$

Poisson Distribution

Properties:

2) $E[X] = \lambda$

$$\begin{aligned} E[X] &= e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \end{aligned}$$

Poisson Distribution

Properties:

2) $E[X] = \lambda$

$$
\begin{aligned}
E[X] &= e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} \\
&= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}
\end{aligned}
$$

Define $j = k - 1$, then

Poisson Distribution

Properties:

2) $E[X] = \lambda$

$$E[X] = e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!}$$
$$= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

Define $j = k - 1$, then

$$E[X] = e^{-\lambda} \lambda \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Poisson Distribution

Properties:

2) $E[X] = \lambda$

$$
\begin{aligned}
E[X] &= e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} \\
&= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}
\end{aligned}
$$

Define $j = k - 1$, then

$$
\begin{aligned}
E[X] &= e^{-\lambda} \lambda \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\
&= e^{-\lambda} \lambda e^{\lambda}
\end{aligned}
$$

Poisson Distribution

Properties:

2) $E[X] = \lambda$

$$\begin{aligned} E[X] &= e^{-\lambda}\sum_{k=0}^{\infty} k\frac{\lambda^k}{k!} \\ &= e^{-\lambda}\lambda\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} \end{aligned}$$

Define $j = k - 1$, then

$$\begin{aligned} E[X] &= e^{-\lambda}\lambda\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} \\ &= e^{-\lambda}\lambda e^{\lambda} \\ &= \lambda \end{aligned}$$

Poisson Distribution

Properties:

Poisson Distribution

Properties:

3) $\text{var}(X) = \lambda$

Poisson Distribution

Properties:

3) $\text{var}(X) = \lambda$

$$E[X^2] = \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda} \lambda^k}{k!}$$

Poisson Distribution

Properties:

3) $\mathrm{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda}\lambda^k}{k!} \\ &= \lambda e^{-\lambda}\left(\sum_{k=1}^{\infty} \frac{k\lambda^{k-1}}{(k-1)!}\right) \end{aligned}$$

Poisson Distribution

Properties:

3) $\text{var}(X) = \lambda$

$$
\begin{aligned}
E[X^2] &= \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda} \lambda^k}{k!} \\
&= \lambda e^{-\lambda} \left(\sum_{k=1}^{\infty} \frac{k \lambda^{k-1}}{(k-1)!} \right)
\end{aligned}
$$

Let $j = k - 1$,

Poisson Distribution

Properties:

3) $\text{var}(X) = \lambda$

$$E[X^2] = \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda} \lambda^k}{k!}$$

$$= \lambda e^{-\lambda} \left(\sum_{k=1}^{\infty} \frac{k \lambda^{k-1}}{(k-1)!} \right)$$

Let $j = k - 1$,

$$E[X^2] = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{(j+1)\lambda^j}{j!}$$

Poisson Distribution

Properties:

3) $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda} \lambda^k}{k!} \\ &= \lambda e^{-\lambda} \left(\sum_{k=1}^{\infty} \frac{k \lambda^{k-1}}{(k-1)!} \right) \end{aligned}$$

Let $j = k - 1$,

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{(j+1)\lambda^j}{j!} \\ &= \lambda e^{-\lambda} \left(\sum_{j=0}^{\infty} \frac{(j)\lambda^j}{j!} + \sum_{j=0}^{\infty} \frac{(1)\lambda^j}{j!} \right) \end{aligned}$$

Poisson Distribution

Properties:

3) $\text{var}(X) = \lambda$

$$\begin{aligned}
E[X^2] &= \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda} \lambda^k}{k!} \\
&= \lambda e^{-\lambda} \left(\sum_{k=1}^{\infty} \frac{k \lambda^{k-1}}{(k-1)!} \right)
\end{aligned}$$

Let $j = k - 1$,

$$\begin{aligned}
E[X^2] &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{(j+1)\lambda^j}{j!} \\
&= \lambda e^{-\lambda} \left(\sum_{j=0}^{\infty} \frac{(j)\lambda^j}{j!} + \sum_{j=0}^{\infty} \frac{(1)\lambda^j}{j!} \right) \\
&= \lambda e^{-\lambda} (\lambda e^{\lambda} + e^{\lambda})
\end{aligned}$$

Poisson Distribution

Properties

3) $\text{var}(X) = \lambda$

$$E[X^2] = \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda})$$

Poisson Distribution

Properties

3) $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda}) \\ &= \lambda(\lambda + 1) \end{aligned}$$

Poisson Distribution

Properties

3) $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda}) \\ &= \lambda(\lambda + 1) \end{aligned}$$

$\text{var}(X) = E[X^2] - E[X]$

Poisson Distribution

Properties

3) $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda}) \\ &= \lambda(\lambda + 1) \end{aligned}$$

$\text{var}(X) = E[X^2] - E[X] = \lambda^2 + \lambda - \lambda^2 = \lambda$

Poisson Distribution

Properties

3) $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda}) \\ &= \lambda(\lambda + 1) \end{aligned}$$

$\text{var}(X) = E[X^2] - E[X] = \lambda^2 + \lambda - \lambda^2 = \lambda$
Very useful distribution.

Joint PMFs

If we have more than one random variable, we can study their joint behavior.
If (x, y) are possible values of random variables X and Y, then

$$p_{X,Y}(x, y) = P(X = x,\ Y = y)$$

Joint PMFs

If we have more than one random variable, we can study their joint behavior.
If (x, y) are possible values of random variables X and Y, then

$$p_{X,Y}(x, y) = P(X = x \cap Y = y)$$

Joint PMFs

If we want to know the probability of a set of joint events $(x, y) \in A$

$$P((X, Y) \in A) = \sum_{(x,\ y) \in A} p_{X,Y}(x,\ y)$$

We can also calculate the pmfs of X and Y individually (these are the marginal distributions:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Example of Joint PMF

$$\Omega_1 = \{\text{Non-Democracy}, \quad \text{Democracy}\}$$

$$X(\omega) = \begin{cases} 0 \text{ if } \omega = \text{Non-Democracy} \\ 1 \text{ if } \omega = \text{Democracy} \end{cases}$$

$$\Omega_2 = \{\text{not-Killed}, \quad \text{Killed}\}$$

$$Y(\omega) = \begin{cases} y = 0 \text{ if } \omega = \text{not Killed} \\ y = 1 \text{ if } \omega = \text{Killed} \end{cases}$$

$p_{X,Y}$	0	1	p_Y
1	0.034	0.005	0.039
0	0.466	0.495	0.961
p_X	0.5	0.5	1

Example of Joint PMF

$$\Omega_1 = \{\text{Non-Democracy}, \quad \text{Democracy}\}$$

$$X(\omega) = \begin{cases} 0 \text{ if } \omega = \text{Non-Democracy} \\ 1 \text{ if } \omega = \text{Democracy} \end{cases}$$

$$\Omega_2 = \{\text{not-Killed}, \quad \text{Killed}\}$$

$$Y(\omega) = \begin{cases} y = 0 \text{ if } \omega = \text{not Killed} \\ y = 1 \text{ if } \omega = \text{Killed} \end{cases}$$

$p_{X,Y}$	0	1	p_Y
1	0.034	0.005	0.039
0	0.466	0.495	0.961
p_X	0.5	0.5	1

Example of Joint PMF

$$\Omega_1 = \{\text{Non-Democracy}, \quad \text{Democracy}\}$$

$$X(\omega) = \begin{cases} 0 \text{ if } \omega = \text{Non-Democracy} \\ 1 \text{ if } \omega = \text{Democracy} \end{cases}$$

$$\Omega_2 = \{\text{not-Killed}, \quad \text{Killed}\}$$

$$Y(\omega) = \begin{cases} y = 0 \text{ if } \omega = \text{not Killed} \\ y = 1 \text{ if } \omega = \text{Killed} \end{cases}$$

$p_{X,Y}$	0	1	p_Y
1	0.034	0.005	0.039
0	0.466	0.495	0.961
p_X	0.5	0.5	1

Linearity

$$E[X+Y] = E[X] + E[Y]$$

Proof:

$$E[X+Y] = \sum_x \sum_y (x+y)p(x,y)$$

$$= \sum_x \sum_y xp(x,y) + \sum_x \sum_y yp(x,y)$$

$$= \sum_x x \sum_y p(x,y) + \sum_y y \sum_x p(x,y)$$

$$= \sum_x xp(x) + \sum_y yp(y)$$

Linearity

$$E[X+Y] = E[X] + E[Y]$$

Proof:

$$E[X+Y] = \sum_x \sum_y (x+y)p(x,y)$$

$$= \sum_x \sum_y xp(x,y) + \sum_x \sum_y yp(x,y)$$

$$= \sum_x x \sum_y p(x,y) + \sum_y y \sum_x p(x,y)$$

$$= \sum_x xp(x) + \sum_y yp(y)$$

Linearity

$$E[X + Y] = E[X] + E[Y]$$

Proof:

$$E[X + Y] = \sum_x \sum_y (x + y)p(x, y)$$

$$= \sum_x \sum_y xp(x, y) + \sum_x \sum_y yp(x, y)$$

$$= \sum_x x \sum_y p(x, y) + \sum_y y \sum_x p(x, y)$$

$$= \sum_x xp(x) + \sum_y yp(y)$$

Linearity

$$E[X + Y] = E[X] + E[Y]$$

Proof:

$$E[X + Y] = \sum_x \sum_y (x + y)p(x, y)$$

$$= \sum_x \sum_y xp(x, y) + \sum_x \sum_y yp(x, y)$$

$$= \sum_x x \sum_y p(x, y) + \sum_y y \sum_x p(x, y)$$

$$= \sum_x xp(x) + \sum_y yp(y)$$

Linearity

$$E[X + Y] = E[X] + E[Y]$$

Proof:

$$E[X + Y] = \sum_x \sum_y (x + y)p(x, y)$$

$$= \sum_x \sum_y xp(x, y) + \sum_x \sum_y yp(x, y)$$

$$= \sum_x x \sum_y p(x, y) + \sum_y y \sum_x p(x, y)$$

$$= \sum_x xp(x) + \sum_y yp(y)$$

Conditional Probability Distribution Function

Definition

Suppose X and Y are discrete random variables with joint pdf $p(x, y)$. Then define the conditional probability function $p(x|y)$ as

$$p(x|y) = \frac{p(x, y)}{p_Y(y)}$$

Why Does Marginalization Work?

Begin with discrete case.

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x|Y = y)$ as,

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x|Y = y)$ as,

$$P(X = x|Y = y) \equiv \quad = \quad p(x|y)$$

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x|Y = y)$ as,

$$\begin{aligned} P(X = x|Y = y) \equiv &= p(x|y) \\ &= p(x, y)/p(y) \end{aligned}$$

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x|Y = y)$ as,

$$\begin{aligned} P(X = x|Y = y) \equiv &= p(x|y) \\ &= p(x, y)/p(y) \end{aligned}$$

Then it follows that:

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x | Y = y)$ as,

$$\begin{aligned} P(X = x|Y = y) \equiv \; &= p(x|y) \\ &= p(x, y)/p(y) \end{aligned}$$

Then it follows that:

$$p(x, y) = p(x|y)p(y)$$

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x|Y = y)$ as,

$$\begin{aligned} P(X = x|Y = y) \equiv &= p(x|y) \\ &= p(x, y)/p(y) \end{aligned}$$

Then it follows that:

$$p(x, y) = p(x|y)p(y)$$

Marginalizing over y to get $p(x)$ is then,

Why Does Marginalization Work?

Begin with discrete case.

Consider jointly distributed discrete random variables, X and Y. We'll suppose they have joint pmf,

$$P(X = x, Y = y) = p(x, y)$$

Suppose that the distribution allocates its mass at $x_1, x_2, \ldots, x_M$ and $y_1, y_2, \ldots, y_N$.

Define the conditional mass function $P(X = x|Y = y)$ as,

$$\begin{aligned} P(X = x|Y = y) \equiv &= p(x|y) \\ &= p(x, y)/p(y) \end{aligned}$$

Then it follows that:

$$p(x, y) = p(x|y)p(y)$$

Marginalizing over y to get $p(x)$ is then,

$$p(x_j) = \sum_{i=1}^{N} p(x_j|y_i)p(y_i)$$

A Table

	$Y = 0$	$Y = 1$	
$X = 0$	$p(0,0)$	$p(0, 1)$	$p_X(0)$
$X = 1$	$p(1,0)$	$p(1,1)$	$p_X(1)$
	$p_Y(0)$	$p_Y(1)$	

A Table

	$Y = 0$	$Y = 1$	
$X = 0$	0.01	0.05	?
$X = 1$	0.25	0.69	?
	0.26	0.74	

$$p_X(0) = p(0|y = 0)p(y = 0) + p(0|y = 1)p(y = 1)$$

A Table

	$Y=0$	$Y=1$	
$X=0$	0.01	0.05	?
$X=1$	0.25	0.69	?
	0.26	0.74	

$$
\begin{aligned}
p_X(0) &= p(0|y=0)p(y=0) + p(0|y=1)p(y=1) \\
&= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74
\end{aligned}
$$

A Table

	$Y=0$	$Y=1$	
$X=0$	0.01	0.05	?
$X=1$	0.25	0.69	?
	0.26	0.74	

$$
\begin{aligned}
p_X(0) &= p(0|y=0)p(y=0) + p(0|y=1)p(y=1) \\
&= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74 \\
&= 0.06
\end{aligned}
$$

A Table

	Y = 0	Y = 1	
X = 0	0.01	0.05	?
X = 1	0.25	0.69	?
	0.26	0.74	

$$
\begin{aligned}
p_X(0) &= p(0|y=0)p(y=0) + p(0|y=1)p(y=1) \\
&= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74 \\
&= 0.06
\end{aligned}
$$

$$
p_X(1) = p(1|y=0)p(y=0) + p(1|y=1)p(y=1)
$$

A Table

	Y = 0	Y = 1	
X = 0	0.01	0.05	?
X = 1	0.25	0.69	?
	0.26	0.74	

$$\begin{aligned} p_X(0) &= p(0|y=0)p(y=0) + p(0|y=1)p(y=1) \\ &= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74 \\ &= 0.06 \end{aligned}$$

$$\begin{aligned} p_X(1) &= p(1|y=0)p(y=0) + p(1|y=1)p(y=1) \\ &= \frac{0.25}{0.26} \times 0.26 + \frac{0.69}{0.74} \times 0.74 \end{aligned}$$

A Table

	$Y = 0$	$Y = 1$	
$X = 0$	0.01	0.05	?
$X = 1$	0.25	0.69	?
	0.26	0.74	

$$\begin{aligned} p_X(0) &= p(0|y=0)p(y=0) + p(0|y=1)p(y=1) \\ &= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74 \\ &= 0.06 \end{aligned}$$

$$\begin{aligned} p_X(1) &= p(1|y=0)p(y=0) + p(1|y=1)p(y=1) \\ &= \frac{0.25}{0.26} \times 0.26 + \frac{0.69}{0.74} \times 0.74 \\ &= 0.94 \end{aligned}$$

Definition

Two random variables X and Y are independent if for any two sets of real numbers A and B,

Definition

Two random variables X and Y are independent if for any two sets of real numbers A and B,

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$$

Definition

Two random variables X and Y are independent if for any two sets of real numbers A and B,

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$$

Equivalently we will say X and Y are independent if,

Definition

Two random variables X and Y are independent if for any two sets of real numbers A and B,

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$$

Equivalently we will say X and Y are independent if,

$$p(x, y) = p_X(x)p_Y(y)$$

Definition

Two random variables X and Y are independent if for any two sets of real numbers A and B,

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$$

Equivalently we will say X and Y are independent if,

$$p(x, y) = p_X(x)p_Y(y)$$

If X and Y are not independent, we will say they are dependent

Conditional Distribution

If X and Y are independent, then

$$
\begin{aligned}
p_{X|Y}(x|y) &= \frac{p(x,y)}{p_Y(y)} \\
&= \frac{p_X(x)p_Y(y)}{p_Y(y)} \\
&= p_X(x)
\end{aligned}
$$

In words: the distribution of X does not change as levels of Y change.

Conditional Expectation

$$\mathbb{E}(Y|X) = \sum_y y \cdot p_{Y|X}(y|x)$$

$\mathbb{E}(Y|X)$ is the *best* way to predict Y given X.

Continuous Random Variables

Many times we deal with continuous events:

- Growth rate of an economy.
- Length of a recession.
- Where a ball goes through a soccer net.

We will need continuous random variables to assign numbers to these events.

Continuous Random Variables

Many times we deal with continuous events:

- Growth rate of an economy.
- Length of a recession.
- Where a ball goes through a soccer net.

We will need continuous random variables to assign numbers to these events.

Continuous Random Variables

Many times we deal with continuous events:

- Growth rate of an economy.
- Length of a recession.
- Where a ball goes through a soccer net.

We will need continuous random variables to assign numbers to these events.

Continuous Random Variables

Many times we deal with continuous events:

- Growth rate of an economy.
- Length of a recession.
- Where a ball goes through a soccer net.

We will need continuous random variables to assign numbers to these events.

Probability of a continuous random variable

Definition

If X is a continuous random variable, the probability density function of X is the function $f_X(x)$ that satisfies.

$$F_X(x) = P(X \leq x)$$

$$= P(X \in (-\infty,\ x))$$

$$= \int_{-\infty}^{x} f_X(t)dt \quad x \in \mathbf{R}$$

$$\int_a^b f_X(x)dx = P(X \in (a, b))$$

Probability of a continuous random variable

Definition

If X is a continuous random variable, the probability density function of X is the function $f_X(x)$ that satisfies.

$$F_X(x) = P(X \leq x)$$

$$= P(X \in (-\infty,\ x))$$

$$= \int_{-\infty}^{x} f_X(t)dt \qquad x \in \mathbf{R}$$

$$\int_a^b f_X(x)dx = P(X \in (a, b))$$

Probability of a continuous random variable

Definition

If X is a continuous random variable, the probability density function of X is the function $f_X(x)$ that satisfies.

$$F_X(x) = P(X \leq x)$$

$$= P(X \in (-\infty,\ x))$$

$$= \int_{-\infty}^{x} f_X(t)dt \quad x \in \mathbf{R}$$

$$\int_a^b f_X(x)dx = P(X \in (a, b))$$

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Definition

Cumulative Distribution function. For a continuous random variable X define its cumulative distribution function $F(\cdot)$ as,

$$F(t) \quad = \quad P(X \leq t) = \int_{-\infty}^{t} f(x)dx$$

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Definition

Cumulative Distribution function. For a continuous random variable X define its cumulative distribution function $F(\cdot)$ as,

$$F(t) \;=\; P(X \leq t) = \int_{-\infty}^{t} f(x)dx$$

pdf

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Definition

Cumulative Distribution function. For a continuous random variable X define its cumulative distribution function $F(\cdot)$ as,

$$F(t) \quad = \quad P(X \leq t) = \int_{-\infty}^{t} f(x)dx$$

pdf

cdf

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Definition

Cumulative Distribution function. For a continuous random variable X define its cumulative distribution function $F(\cdot)$ as,

$$F(t) \quad = \quad P(X \leq t) = \int_{-\infty}^{t} f(x)dx$$

$$\text{pdf} \xrightarrow{\text{Integrate}} \text{cdf}$$

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Definition

Cumulative Distribution function. For a continuous random variable X define its cumulative distribution function $F(\cdot)$ as,

$$F(t) \;\; = \;\; P(X \leq t) = \int_{-\infty}^{t} f(x)dx$$

$$\text{pdf} \xrightarrow{\text{Integrate}} \text{cdf}$$

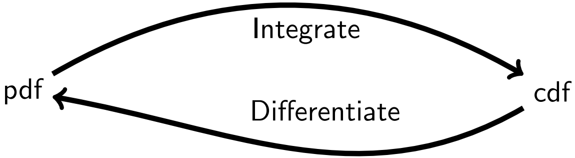
Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

Definition

Cumulative Distribution function. For a continuous random variable X define its cumulative distribution function $F(\cdot)$ as,

$$F(t) = P(X \leq t) = \int_{-\infty}^{t} f(x)dx$$



RV: continuous uniform

Definition

We can define Y to have a uniform distribution on the interval (a, b)

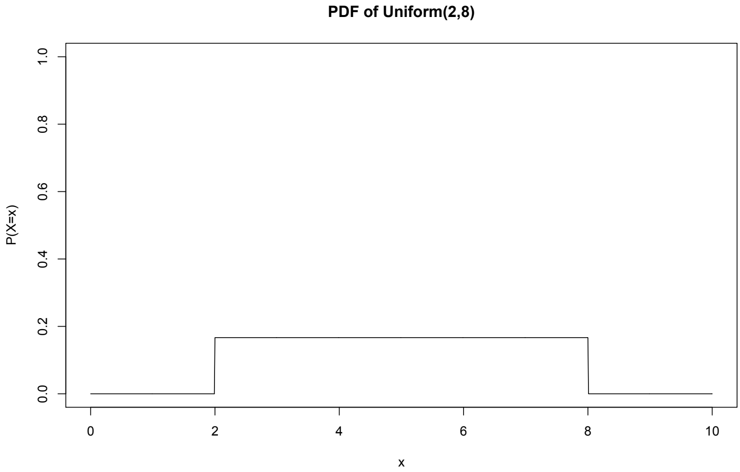
$$f_Y(y) = \begin{cases} \frac{1}{b-a} & \text{if } a < y < b \\ 0 & \text{otherwise} \end{cases}$$

Example: Uniform

Example: Suppose that we are waiting for Comcast to show up and install our cable package. They say that they may arrive between 2:00 and 8:00. Without any further information, you may have no reason to suspect any particular time over any other.

$$X \sim U(2, 8)$$

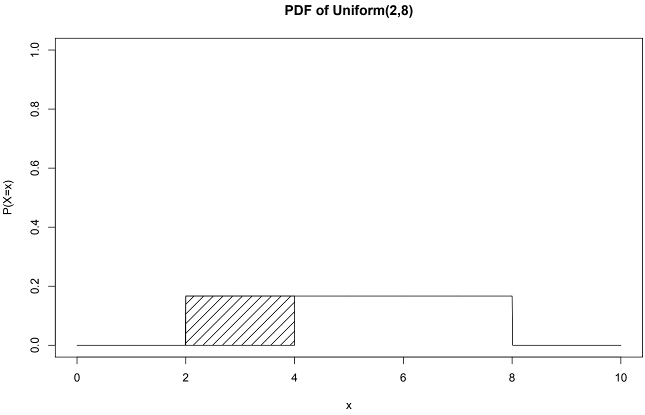
RV: continuous uniform



RV: continuous uniform

What is the probability that the cable installation truck arrives before 4:00?

RV: continuous uniform ($P(X \leq 4)$)



RV: continuous uniform

$$P(Y \leq y) = \int_{x=-\infty}^{y} f(x)dx = \int_{x=-\infty}^{y} \frac{1}{b-a}dx = \int_{x=a}^{y} \frac{1}{b-a}dx$$

$$\frac{y}{b-a} - \frac{a}{b-a} = \frac{y-a}{b-a}$$

$$= \frac{4-2}{8-2}$$

RV: continuous uniform

$$P(Y \leq y) = \int_{x=-\infty}^{y} f(x)dx = \int_{x=-\infty}^{y} \frac{1}{b-a}dx = \int_{x=a}^{y} \frac{1}{b-a}dx$$

$$\frac{y}{b-a} - \frac{a}{b-a} = \frac{y-a}{b-a}$$

$$= \frac{4-2}{8-2}$$

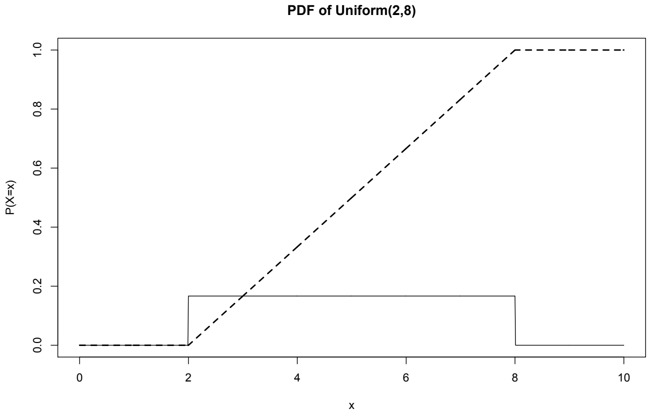
RV: continuous uniform

$$P(Y \leq y) = \int_{x=-\infty}^{y} f(x)dx = \int_{x=-\infty}^{y} \frac{1}{b-a}dx = \int_{x=a}^{y} \frac{1}{b-a}dx$$

$$\frac{y}{b-a} - \frac{a}{b-a} = \frac{y-a}{b-a}$$

$$= \frac{4-2}{8-2}$$

RV: continuous uniform



Common Continuous Distributions: Exponential

Example: Suppose that the average durability of a Peace Accord in the Democratic Republic of Congo is 6 months. How can we model the time to failure of Peace Accord?

The exponential distribution is often used to model 'time to failure' processes.

$X \sim \exp(\lambda)$ if its PDF is:

$$f(x) = \lambda e^{-\lambda x}$$

with $f(x) = 0$ for $x < 0$

λ is known as the 'rate' parameter.

$\lambda = \frac{1}{E(X)}$

Common Continuous Distributions: Exponential

Example: Suppose that the average durability of a Peace Accord in the Democratic Republic of Congo is 6 months. How can we model the time to failure of Peace Accord?
The exponential distribution is often used to model 'time to failure' processes.
$X \sim \exp(\lambda)$ if its PDF is:

$$f(x) = \lambda e^{-\lambda x}$$

with $f(x) = 0$ for $x < 0$
λ is known as the 'rate' parameter.
$\lambda = \frac{1}{E(X)}$

Common Continuous Distributions: Exponential

Example: Suppose that the average durability of a Peace Accord in the Democratic Republic of Congo is 6 months. How can we model the time to failure of Peace Accord?

The exponential distribution is often used to model 'time to failure' processes.

$X \sim \exp(\lambda)$ if its PDF is:

$$f(x) = \lambda e^{-\lambda x}$$

with $f(x) = 0$ for $x < 0$

λ is known as the 'rate' parameter.

$\lambda = \frac{1}{E(X)}$

Common Continuous Distributions: Exponential

Example: Suppose that the average durability of a Peace Accord in the Democratic Republic of Congo is 6 months. How can we model the time to failure of Peace Accord?

The exponential distribution is often used to model 'time to failure' processes.

$X \sim \exp(\lambda)$ if its PDF is:

$$f(x) = \lambda e^{-\lambda x}$$

with $f(x) = 0$ for $x < 0$

λ is known as the 'rate' parameter.

$\lambda = \frac{1}{E(X)}$

Exponential CDF

The CDF of $X \sim \exp(\lambda)$ is:

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{0}^{x} \lambda e^{-\lambda x} dx$$

$$= F(x) - F(0)$$

where F(x) is the antiderivative of f(x).

$$= -\frac{\lambda e^{-\lambda x}}{\lambda} - \left(-\frac{\lambda e^{-\lambda * 0}}{\lambda}\right)$$

$$= 1 - e^{-\lambda x}$$

Exponential CDF

The CDF of $X \sim \exp(\lambda)$ is:

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{0}^{x} \lambda e^{-\lambda x} dx$$

$$= F(x) - F(0)$$

where F(x) is the antiderivative of f(x).

$$= -\frac{\lambda e^{-\lambda x}}{\lambda} - \left(-\frac{\lambda e^{-\lambda * 0}}{\lambda}\right)$$

$$= 1 - e^{-\lambda x}$$

Exponential CDF

The CDF of $X \sim \exp(\lambda)$ is:

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{0}^{x} \lambda e^{-\lambda x} dx$$

$$= F(x) - F(0)$$

where F(x) is the antiderivative of f(x).

$$= -\frac{\lambda e^{-\lambda x}}{\lambda} - (-\frac{\lambda e^{-\lambda * 0}}{\lambda})$$

$$= 1 - e^{-\lambda x}$$

Exponential CDF

The CDF of $X \sim \exp(\lambda)$ is:

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{0}^{x} \lambda e^{-\lambda x} dx$$

$$= F(x) - F(0)$$

where F(x) is the antiderivative of f(x).

$$= -\frac{\lambda e^{-\lambda x}}{\lambda} - (-\frac{\lambda e^{-\lambda * 0}}{\lambda})$$

$$= 1 - e^{-\lambda x}$$

Exponential CDF

The CDF of $X \sim \exp(\lambda)$ is:

$$F(x) = \int_{-\infty}^{x} f(x)dx = \int_{0}^{x} \lambda e^{-\lambda x} dx$$

$$= F(x) - F(0)$$

where F(x) is the antiderivative of f(x).

$$= -\frac{\lambda e^{-\lambda x}}{\lambda} - (-\frac{\lambda e^{-\lambda * 0}}{\lambda})$$

$$= 1 - e^{-\lambda x}$$

Expectation With Continuous Random Variables

Definition

If X is a continuous random variable then,

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

Suppose $X \sim Uniform(0, 1)$. What is $E[X]$?

Suppose $X \sim Uniform(0, 1)$. What is $E[X]$?

$$E[X]$$

Suppose $X \sim \mathit{Uniform}(0, 1)$. What is $E[X]$?

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Suppose $X \sim Uniform(0,1)$. What is $E[X]$?

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_{-\infty}^{0} x0dx + \int_{0}^{1} x1dx + \int_{1}^{\infty} x0dx \end{aligned}$$

Suppose $X \sim Uniform(0, 1)$. What is $E[X]$?

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_{-\infty}^{0} x0dx + \int_{0}^{1} x1dx + \int_{1}^{\infty} x0dx \\ &= 0 + \frac{x^2}{2}\Big|_0^1 + 0 \end{aligned}$$

Suppose $X \sim \mathit{Uniform}(0, 1)$. What is $E[X]$?

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_{-\infty}^{0} x 0 dx + \int_{0}^{1} x 1 dx + \int_{1}^{\infty} x 0 dx \\ &= 0 + \frac{x^2}{2}\Big|_0^1 + 0 \\ &= 0 + \frac{1}{2} + 0 \end{aligned}$$

Suppose $X \sim \mathit{Uniform}(0,1)$. What is $E[X]$?

$$
\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x f(x) dx \\
&= \int_{-\infty}^{0} x0 dx + \int_{0}^{1} x1 dx + \int_{1}^{\infty} x0 dx \\
&= 0 + \frac{x^2}{2}\Big|_0^1 + 0 \\
&= 0 + \frac{1}{2} + 0 \\
&= \frac{1}{2}
\end{aligned}
$$

Expectations of Functions

Proposition

Suppose X is a continuous random variable and $g : \Re \to \Re$. Then,

$$E[g(X)] \;=\; \int_{-\infty}^{\infty} g(x)f(x)dx$$

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0, 1)$. What is E[g(X)]?

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0, 1)$. What is E[g(X)]?

$$E[g(X)]$$

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0,1)$. What is E[g(X)]?

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0, 1)$. What is E[g(X)]?

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x) f(x) dx \\ &= \int_0^1 x^2 \frac{1}{1-0} dx \end{aligned}$$

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0,1)$. What is E[g(X)]?

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x) f(x) dx \\ &= \int_0^1 x^2 \frac{1}{1-0} dx \\ &= \frac{x^3}{3}\Big|_0^1 \end{aligned}$$

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0,1)$. What is E[g(X)]?

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x) f(x) dx \\ &= \int_0^1 x^2 \frac{1}{1-0} dx \\ &= \frac{x^3}{3} \Big|_0^1 \\ &= \frac{1}{3} \end{aligned}$$

Corollary

Suppose X is a continuous random variable. Then,

$$E[aX + b] = aE[X] + b$$

Proof.

Corollary

Suppose X is a continuous random variable. Then,

$$E[aX + b] \;\; = \;\; aE[X] + b$$

Proof.

$$E[aX + b] \;\; = \;\; \int_{-\infty}^{\infty} (ax + b) f(x) dx$$

Corollary

Suppose X is a continuous random variable. Then,

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b) f(x) dx \\ &= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx \end{aligned}$$

□

Corollary

Suppose X is a continuous random variable. Then,

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b)f(x)dx \\ &= a\int_{-\infty}^{\infty} xf(x)dx + b\int_{-\infty}^{\infty} f(x)dx \\ &= aE[X] + b \times 1 \end{aligned}$$

□

Definition

Variance. If X is a continuous random variable, define its variance, $Var(X)$,

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] \\ &= \int_{-\infty}^{\infty} (x - E[X])^2 f(x) dx \\ &= E[X^2] - E[X]^2 \end{aligned}$$

Variance: Random Variable

$X \sim \text{Uniform}(0, 1)$. What is $Var(X)$?

Variance: Random Variable

$X \sim \text{Uniform}(0,1)$. What is $Var(X)$?

$$E[X^2] \;=\; \frac{1}{3}$$

Variance: Random Variable

$X \sim \text{Uniform}(0, 1)$. What is $Var(X)$?

$$\begin{aligned} E[X^2] &= \frac{1}{3} \\ E[X]^2 &= \left(\frac{1}{2}\right)^2 \end{aligned}$$

Variance: Random Variable

$X \sim \text{Uniform}(0,1)$. What is $Var(X)$?

$$\begin{aligned} E[X^2] &= \frac{1}{3} \\ E[X]^2 &= \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{4} \end{aligned}$$

Variance: Random Variable

$X \sim \text{Uniform}(0, 1)$. What is $Var(X)$?

$$\begin{aligned} E[X^2] &= \frac{1}{3} \\ E[X]^2 &= \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{4} \end{aligned}$$

$$Var(X) = E[X^2] - E[X]^2$$

Variance: Random Variable

$X \sim \text{Uniform}(0, 1)$. What is $Var(X)$?

$$\begin{aligned} E[X^2] &= \frac{1}{3} \\ E[X]^2 &= \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} Var(X) &= E[X^2] - E[X]^2 \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

Definition

Suppose X is a random variable with $X \in \mathbf{R}$ and probability density function

$$f(x) \quad = \quad \frac{1}{\sqrt{2\sigma^2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Then X is a normally distributed random variable with parameters μ and σ^2.
Equivalently, we'll write

$$X \quad \sim \quad \textit{Normal}(\mu, \sigma^2)$$

Support for President Trump

Suppose we are interested in modeling presidential approval

Support for President Trump

Suppose we are interested in modeling presidential approval

- Let Y represent random variable: proportion of population who "approves job president is doing"

Support for President Trump

Suppose we are interested in modeling presidential approval

- Let Y represent random variable: proportion of population who "approves job president is doing"
- Individual responses (that constitute proportion) are independent and identically distributed (sufficient, not necessary) and we take the average of those individual responses

Support for President Trump

Suppose we are interested in modeling presidential approval

- Let Y represent random variable: proportion of population who "approves job president is doing"
- Individual responses (that constitute proportion) are independent and identically distributed (sufficient, not necessary) and we take the average of those individual responses
- Observe many responses ($N \to \infty$)

Support for President Trump

Suppose we are interested in modeling presidential approval

- Let Y represent random variable: proportion of population who "approves job president is doing"
- Individual responses (that constitute proportion) are independent and identically distributed (sufficient, not necessary) and we take the average of those individual responses
- Observe many responses ($N \rightarrow \infty$)
- Then (by Central Limit Theorem) Y is Normally distributed, or

Support for President Trump

Suppose we are interested in modeling presidential approval

- Let Y represent random variable: proportion of population who "approves job president is doing"
- Individual responses (that constitute proportion) are independent and identically distributed (sufficient, not necessary) and we take the average of those individual responses
- Observe many responses ($N \to \infty$)
- Then (by Central Limit Theorem) Y is Normally distributed, or

$$Y \quad \sim \quad \text{Normal}(\mu, \sigma^2)$$

Support for President Trump

Suppose we are interested in modeling presidential approval

- Let Y represent random variable: proportion of population who "approves job president is doing"
- Individual responses (that constitute proportion) are independent and identically distributed (sufficient, not necessary) and we take the average of those individual responses
- Observe many responses ($N \rightarrow \infty$)
- Then (by Central Limit Theorem) Y is Normally distributed, or

$$\begin{aligned} Y &\sim \text{Normal}(\mu, \sigma^2) \\ f(y) &= \frac{\exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} \end{aligned}$$

Expected Value/Variance of Normal Distribution

Z is a standard normal distribution if

Expected Value/Variance of Normal Distribution

Z is a standard normal distribution if

$$Z \sim \text{Normal}(0, 1)$$

Expected Value/Variance of Normal Distribution

Z is a standard normal distribution if

$$Z \sim \text{Normal}(0, 1)$$

We'll call the cumulative density function of Z,

Expected Value/Variance of Normal Distribution

Z is a standard normal distribution if

$$Z \sim \text{Normal}(0, 1)$$

We'll call the cumulative density function of Z,

$$F_Z(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp(-z^2/2) dz$$

Expected Value/Variance of Normal Distribution

Z is a standard normal distribution if

$$Z \sim \text{Normal}(0, 1)$$

We'll call the cumulative density function of Z,

$$F_Z(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp(-z^2/2) dz$$

Proposition

Scale/Location. If $Z \sim N(0, 1)$, then $X = aZ + b$ is,

$$X \sim Normal(b, a^2)$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.
Call $F_Z(x)$ cdf for standardized normal.

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.
Call $F_Z(x)$ cdf for standardized normal.

$$F_Y(x) = P(Y \leq x)$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.
Call $F_Z(x)$ cdf for standardized normal.

$$\begin{aligned} F_Y(x) &= P(Y \leq x) \\ &= P(aZ + b \leq x) \end{aligned}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.
Call $F_Z(x)$ cdf for standardized normal.

$$\begin{aligned} F_Y(x) &= P(Y \leq x) \\ &= P(aZ + b \leq x) \\ &= P(Z \leq \left[\frac{x-b}{a}\right]) \end{aligned}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.
Call $F_Z(x)$ cdf for standardized normal.

$$\begin{aligned}
F_Y(x) &= P(Y \leq x) \\
&= P(aZ + b \leq x) \\
&= P(Z \leq \left[\frac{x-b}{a}\right]) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-b}{a}} \exp(-\frac{z^2}{2}) dz
\end{aligned}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

To prove we need to show that density for Y is a normal distribution.
That is, we'll show $F_Y(x)$ is Normal cdf.
Call $F_Z(x)$ cdf for standardized normal.

$$\begin{aligned}
F_Y(x) &= P(Y \leq x) \\
&= P(aZ + b \leq x) \\
&= P(Z \leq \left[\frac{x-b}{a}\right]) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-b}{a}} \exp(-\frac{z^2}{2}) dz \\
&= F_Z(\frac{x-b}{a})
\end{aligned}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

$$\frac{\partial F_Y(x)}{\partial x} = \frac{\partial F_Z(\frac{x-b}{a})}{\partial x}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

$$\begin{aligned}
\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\
&= f_Z(\frac{x-b}{a})\frac{1}{a} \text{ By the chain rule}
\end{aligned}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

$$\begin{aligned}
\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\
&= f_Z(\frac{x-b}{a})\frac{1}{a} \text{ By the chain rule} \\
&= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{\left(\frac{x-b}{a}\right)^2}{2}\right] \text{ By definition of } f_Z(x)
\end{aligned}$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

$$
\begin{aligned}
\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\
&= f_Z(\frac{x-b}{a})\frac{1}{a} \text{ By the chain rule} \\
&= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{\left(\frac{x-b}{a}\right)^2}{2}\right] \text{ By definition of } f_Z(x) \\
&= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(x-b)^2}{2a^2}\right]
\end{aligned}
$$

Proof: $Z \sim N(0,1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

$$
\begin{aligned}
\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\
&= f_Z(\frac{x-b}{a})\frac{1}{a} \text{ By the chain rule} \\
&= \frac{1}{\sqrt{2\pi}a}\exp\left[-\frac{\left(\frac{x-b}{a}\right)^2}{2}\right] \text{ By definition of } f_Z(x) \\
&= \frac{1}{\sqrt{2\pi}a}\exp\left[-\frac{(x-b)^2}{2a^2}\right] \\
&= \text{Normal}(b, a^2)
\end{aligned}
$$

Expectation and Variance

Assume we know:

$$
\begin{aligned}
E[Z] &= 0 \\
Var(Z) &= 1
\end{aligned}
$$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$E[Y] = E[\sigma Z + \mu]$$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned} E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \end{aligned}$$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned} E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \\ &= \mu \end{aligned}$$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned} E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \\ &= \mu \\ Var(Y) &= Var(\sigma Z + \mu) \end{aligned}$$

Expectation and Variance

Assume we know:

$$
\begin{aligned}
E[Z] &= 0 \\
Var(Z) &= 1
\end{aligned}
$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$
\begin{aligned}
E[Y] &= E[\sigma Z + \mu] \\
&= \sigma E[Z] + \mu \\
&= \mu \\
Var(Y) &= Var(\sigma Z + \mu) \\
&= \sigma^2 Var(Z) + Var(\mu)
\end{aligned}
$$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned} E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \\ &= \mu \\ Var(Y) &= Var(\sigma Z + \mu) \\ &= \sigma^2 Var(Z) + Var(\mu) \\ &= \sigma^2 + 0 \end{aligned}$$

Expectation and Variance

Assume we know:

$$\begin{aligned} E[Z] &= 0 \\ Var(Z) &= 1 \end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned} E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \\ &= \mu \\ Var(Y) &= Var(\sigma Z + \mu) \\ &= \sigma^2 Var(Z) + Var(\mu) \\ &= \sigma^2 + 0 \\ &= \sigma^2 \end{aligned}$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

$$P(Y \geq 0.45) \;\; = \;\; 1 - P(Y \leq 0.45)$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

$$\begin{aligned} P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\ &= 1 - P(0.05Z + 0.39 \leq 0.45) \end{aligned}$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

$$\begin{aligned}
P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\
&= 1 - P(0.05Z + 0.39 \leq 0.45) \\
&= 1 - P(Z \leq \frac{0.45 - 0.39}{0.05})
\end{aligned}$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

$$
\begin{aligned}
P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\
&= 1 - P(0.05Z + 0.39 \leq 0.45) \\
&= 1 - P(Z \leq \frac{0.45 - 0.39}{0.05}) \\
&= 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{6/5} \exp(-z^2/2) dz
\end{aligned}
$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

$$
\begin{aligned}
P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\
&= 1 - P(0.05Z + 0.39 \leq 0.45) \\
&= 1 - P(Z \leq \frac{0.45 - 0.39}{0.05}) \\
&= 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{6/5} \exp(-z^2/2) dz \\
&= 1 - F_Z(\frac{6}{5})
\end{aligned}
$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$
$P(Y \geq 0.45)$

$$\begin{aligned}
P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\
&= 1 - P(0.05Z + 0.39 \leq 0.45) \\
&= 1 - P(Z \leq \frac{0.45 - 0.39}{0.05}) \\
&= 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{6/5} \exp(-z^2/2) dz \\
&= 1 - F_Z(\frac{6}{5}) \\
&= 0.1150697
\end{aligned}$$

The Gamma Function

Definition

Suppose $\alpha > 0$*. Then define* $\Gamma(\alpha)$ *as*

$$\Gamma(\alpha) \quad = \quad \int_0^\infty y^{\alpha-1} e^{-y} dy$$

- For $\alpha \in \{1, 2, 3, \ldots\}$
  $\Gamma(\alpha) = (\alpha - 1)!$
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned} \frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1} e^{-y} dy}{\Gamma(\alpha)} \\ 1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1} e^{-y} dy \end{aligned}$$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned} \frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1} e^{-y} dy}{\Gamma(\alpha)} \\ 1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1} e^{-y} dy \end{aligned}$$

Set $X = Y/\beta$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned}\frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1}e^{-y}dy}{\Gamma(\alpha)} \\ 1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1}e^{-y}dy\end{aligned}$$

Set $X = Y/\beta$

$$F(x) = P(X \leq x) = P(Y/\beta \leq x)$$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned}
\frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1}e^{-y}dy}{\Gamma(\alpha)} \\
1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1}e^{-y}dy
\end{aligned}$$

Set $X = Y/\beta$

$$\begin{aligned}
F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\
&= P(Y \leq x\beta)
\end{aligned}$$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned}\frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1}e^{-y}dy}{\Gamma(\alpha)} \\ 1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1}e^{-y}dy\end{aligned}$$

Set $X = Y/\beta$

$$\begin{aligned}F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\ &= P(Y \leq x\beta) \\ &= F_Y(x\beta)\end{aligned}$$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned}
\frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1}e^{-y}dy}{\Gamma(\alpha)} \\
1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1}e^{-y}dy
\end{aligned}$$

Set $X = Y/\beta$

$$\begin{aligned}
F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\
&= P(Y \leq x\beta) \\
&= F_Y(x\beta) \\
\frac{\partial F_Y(x\beta)}{\partial x} &= f_Y(x\beta)\beta
\end{aligned}$$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned}\frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1}e^{-y}dy}{\Gamma(\alpha)}\\ 1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)}y^{\alpha-1}e^{-y}dy\end{aligned}$$

Set $X = Y/\beta$

$$\begin{aligned}F(x) = P(X \leq x) &= P(Y/\beta \leq x)\\ &= P(Y \leq x\beta)\\ &= F_Y(x\beta)\\ \frac{\partial F_Y(x\beta)}{\partial x} &= f_Y(x\beta)\beta\end{aligned}$$

The result is:

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

$$\begin{aligned}
\frac{\Gamma(\alpha)}{\Gamma(\alpha)} &= \frac{\int_0^\infty y^{\alpha-1}e^{-y}dy}{\Gamma(\alpha)} \\
1 &= \int_0^\infty \frac{1}{\Gamma(\alpha)}y^{\alpha-1}e^{-y}dy
\end{aligned}$$

Set $X = Y/\beta$

$$\begin{aligned}
F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\
&= P(Y \leq x\beta) \\
&= F_Y(x\beta) \\
\frac{\partial F_Y(x\beta)}{\partial x} &= f_Y(x\beta)\beta
\end{aligned}$$

The result is:

$$f(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-x\beta}$$

Definition

Suppose X is a continuous random variable, with $X \geq 0$. Then if the pdf of X is

$$f(x|\alpha, \beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\beta}$$

if $x \geq 0$ and 0 otherwise, we will say X is a Gamma distribution.

$$X \sim Gamma(\alpha, \beta)$$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$E[X] = \frac{\alpha}{\beta}$$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned}$$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned}$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned}$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

$$X \sim \text{Gamma}(1, \lambda)$$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$
\begin{aligned}
E[X] &= \frac{\alpha}{\beta} \\
\text{var}(X) &= \frac{\alpha}{\beta^2}
\end{aligned}
$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

$$
\begin{aligned}
X &\sim \text{Gamma}(1, \lambda) \\
f(x|1, \lambda) &= \lambda e^{-x\lambda}
\end{aligned}
$$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned}$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

$$\begin{aligned} X &\sim \text{Gamma}(1, \lambda) \\ f(x|1, \lambda) &= \lambda e^{-x\lambda} \end{aligned}$$

We will say

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned}$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

$$\begin{aligned} X &\sim \text{Gamma}(1, \lambda) \\ f(x|1, \lambda) &= \lambda e^{-x\lambda} \end{aligned}$$

We will say

$$X \sim \text{Exponential}(\lambda)$$

Properties of Gamma Distributions

Proposition

Suppose we have a sequence of independent random variables, with

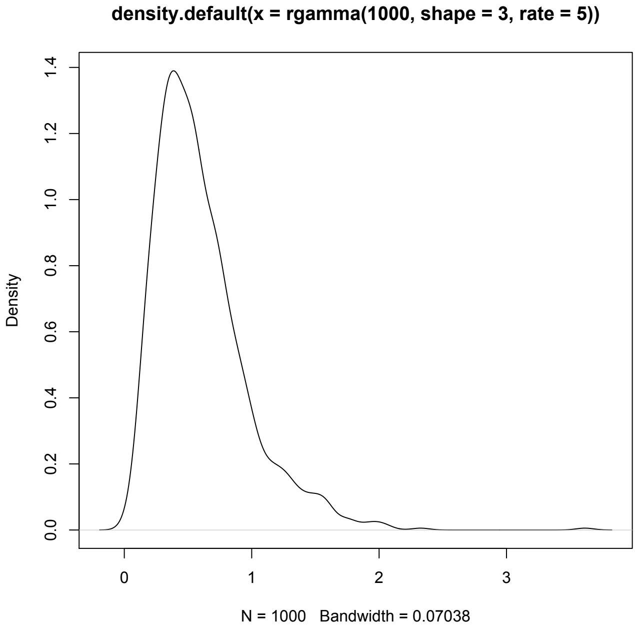
$$X_i \quad \sim \quad Gamma(\alpha_i, \beta)$$

Then

$$Y = \sum_{i=1}^{N} X_i$$

$Y \sim Gamma(\sum_{i=1}^{N} \alpha_i, \beta)$

We can evaluate in `R` with `dgamma` and simulate with `rgamma`
$X \sim \text{Gamma}(3, 5)$ and we evaluate at 3,
`dgamma(3, shape= 3, rate = 5)`
and we can simulate with
`rgamma(1000, shape = 3, rate = 5)`


density.default(x = rgamma(1000, shape = 3, rate = 5))
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
1
2
3
N = 1000 Bandwidth = 0.07038

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

$$F_X(x) = P(X \leq x)$$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(Z^2 \leq x) \end{aligned}$$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(Z^2 \leq x) \\ &= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \end{aligned}$$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0,1)$.
Consider $X = Z^2$

$$
\begin{aligned}
F_X(x) &= P(X \leq x) \\
&= P(Z^2 \leq x) \\
&= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{x}}^{\sqrt{x}} e^{-\frac{z^2}{2}} dz
\end{aligned}
$$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

$$
\begin{aligned}
F_X(x) &= P(X \leq x) \\
&= P(Z^2 \leq x) \\
&= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{x}}^{\sqrt{x}} e^{-\frac{z^2}{2}} dz \\
&= F_Z(\sqrt{x}) - F_Z(-\sqrt{x})
\end{aligned}
$$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

$$
\begin{aligned}
F_X(x) &= P(X \leq x) \\
&= P(Z^2 \leq x) \\
&= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{x}}^{\sqrt{x}} e^{-\frac{z^2}{2}} dz \\
&= F_Z(\sqrt{x}) - F_Z(-\sqrt{x})
\end{aligned}
$$

The pdf then is

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.
Consider $X = Z^2$

$$\begin{aligned}
F_X(x) &= P(X \leq x) \\
&= P(Z^2 \leq x) \\
&= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{x}}^{\sqrt{x}} e^{-\frac{z^2}{2}} dz \\
&= F_Z(\sqrt{x}) - F_Z(-\sqrt{x})
\end{aligned}$$

The pdf then is

$$\frac{\partial F_X(x)}{\partial x} = f_Z(\sqrt{x}) \frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x}) \frac{1}{2\sqrt{x}}$$

χ^2 Distribution

$$\frac{\partial F_X(x)}{\partial x} = f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}}$$

χ^2 Distribution

$$
\begin{aligned}
\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\
&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}})
\end{aligned}
$$

χ^2 Distribution

$$
\begin{aligned}
\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\
&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}}) \\
&= \frac{1}{\sqrt{x}}\frac{1}{\sqrt{2\pi}}(e^{-\frac{x}{2}})
\end{aligned}
$$

χ^2 Distribution

$$
\begin{aligned}
\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\
&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}}) \\
&= \frac{1}{\sqrt{x}}\frac{1}{\sqrt{2\pi}}(e^{-\frac{x}{2}}) \\
&= \frac{(\frac{1}{2})^{1/2}}{\Gamma(\frac{1}{2})}\left(x^{1/2-1}e^{-\frac{x}{2}}\right)
\end{aligned}
$$

χ^2 Distribution

$$
\begin{aligned}
\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\
&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}}) \\
&= \frac{1}{\sqrt{x}}\frac{1}{\sqrt{2\pi}}(e^{-\frac{x}{2}}) \\
&= \frac{(\frac{1}{2})^{1/2}}{\Gamma(\frac{1}{2})}\left(x^{1/2-1}e^{-\frac{x}{2}}\right)
\end{aligned}
$$

$X \sim \text{Gamma}(1/2, 1/2)$

χ^2 Distribution

$$
\begin{aligned}
\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\
&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}}) \\
&= \frac{1}{\sqrt{x}}\frac{1}{\sqrt{2\pi}}(e^{-\frac{x}{2}}) \\
&= \frac{(\frac{1}{2})^{1/2}}{\Gamma(\frac{1}{2})}\left(x^{1/2-1}e^{-\frac{x}{2}}\right)
\end{aligned}
$$

$X \sim \text{Gamma}(1/2, 1/2)$
Then if $X = \sum_{i=1}^{N} Z^2$

χ^2 Distribution

$$
\begin{aligned}
\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\
&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}}) \\
&= \frac{1}{\sqrt{x}}\frac{1}{\sqrt{2\pi}}(e^{-\frac{x}{2}}) \\
&= \frac{(\frac{1}{2})^{1/2}}{\Gamma(\frac{1}{2})}\left(x^{1/2-1}e^{-\frac{x}{2}}\right)
\end{aligned}
$$

$X \sim \text{Gamma}(1/2, 1/2)$
Then if $X = \sum_{i=1}^{N} Z^2$
$X \sim \text{Gamma}(n/2, 1/2)$

Definition

Suppose X is a continuous random variable with $X \geq 0$, with pdf

$$f(x) \;=\; \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}$$

Then we will say X is a χ^2 distribution with n degrees of freedom. Equivalently,

$$X \;\sim\; \chi^2(n)$$

χ^2 Properties

Suppose $X \sim \chi^2(n)$

$$\begin{aligned}
E[X] &= E\left[\sum_{i=1}^{N} Z_i^2\right] \\
&= \sum_{i=1}^{N} E[Z_i^2] \\
\text{var}(Z_i) &= E[Z_i^2] - E[Z_i]^2 \\
1 &= E[Z_i^2] - 0 \\
E[X] &= n
\end{aligned}$$

χ^2 Properties

$$
\begin{aligned}
\text{var}(X) &= \sum_{i=1}^{N} \text{var}(Z_i^2) \\
&= \sum_{i=1}^{N} \left(E[Z_i^4] - E[Z_i]^2 \right) \\
&= \sum_{i=1}^{N} (3-1) = 2n
\end{aligned}
$$

We will use the χ^2 in all sorts of statistics courses

Student's t-Distribution

Definition

Suppose $Z \sim Normal(0, 1)$ and $U \sim \chi^2(n)$. Define the random variable Y as,

$$Y = \frac{Z}{\sqrt{\frac{U}{n}}}$$

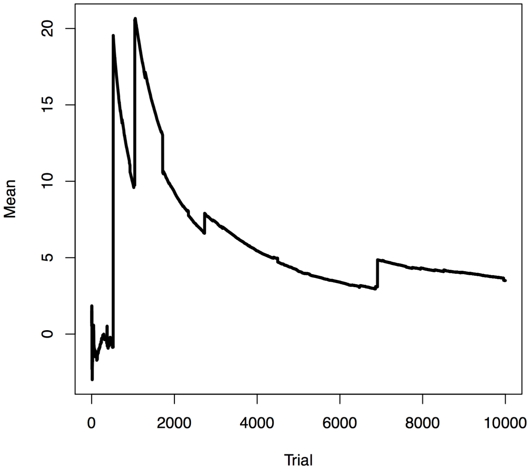
If Z and U are independent then $Y \sim t(n)$, with pdf

$$f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n}\Gamma(\frac{n}{2})}\left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}$$

We will use the t-distribution extensively for test-statistics

Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution



Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution
If $X \sim$ Cauchy(1), then:

Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution
If $X \sim$ Cauchy(1), then:
$E[X] =$ undefined

Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution
If $X \sim$ Cauchy(1), then:
$E[X] =$ undefined
var$(X) =$ undefined

Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution
If $X \sim \text{Cauchy}(1)$, then:
$E[X] = \text{undefined}$
$\text{var}(X) = \text{undefined}$
If $X \sim t(2)$

Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution
If $X \sim \text{Cauchy}(1)$, then:
$E[X] = \text{undefined}$
$\text{var}(X) = \text{undefined}$
If $X \sim t(2)$
$\text{E[X]} = 0$

Student's t-Distribution, Properties

Suppose $n = 1$, Cauchy distribution
If $X \sim \text{Cauchy}(1)$, then:
$E[X] = \text{undefined}$
$\text{var}(X) = \text{undefined}$
If $X \sim t(2)$
$\mathrm{E[X]} = 0$
$\text{var}(X) = \text{undefined}$

Student's t-Distribution, Properties

Suppose $n > 2$, then
$var(X) = \frac{n}{n-2}$
As $n \to \infty$ var$(X) \to 1$.

Using the t-Distribution

We often construct test-statistics
Suppose we take N iid draws,

$$X_i \sim \text{Normal}(\mu, \sigma^2)$$

Define our data set $\boldsymbol{x} = (x_1, \ldots, x_N)$
Calculate:

$$\bar{x} = \sum_{i=1}^{N} \frac{x_i}{N}$$

$$s^2 = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called ‘statistics’.
- If that function is used to infer the value of a parameter of a distribution, it is called an ‘estimator’.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)
$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).
Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called ‘statistics’.
- If that function is used to infer the value of a parameter of a distribution, it is called an ‘estimator’.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)
$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).
Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called ‘statistics’.
- If that function is used to infer the value of a parameter of a distribution, it is called an ‘estimator’.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)
$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).
Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called 'statistics'.
- If that function is used to infer the value of a parameter of a distribution, it is called an 'estimator'.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)
$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).
Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called 'statistics'.
- If that function is used to infer the value of a parameter of a distribution, it is called an 'estimator'.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)

$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).

Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called 'statistics'.
- If that function is used to infer the value of a parameter of a distribution, it is called an 'estimator'.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)
$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).
Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called 'statistics'.
- If that function is used to infer the value of a parameter of a distribution, it is called an 'estimator'.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)
$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).
Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Because estimators are random variables, they have a distribution.
- Our knowledge about the underlying world that produced our data is governed by these distributions.

Estimators

- Because estimators are random variables, they have a distribution.
- Our knowledge about the underlying world that produced our data is governed by these distributions.

Using the T-Distribution

Define

$$
\begin{aligned}
t &= \frac{\bar{x} - \mu}{s/\sqrt{N}} \\
t &\sim t(N-1)
\end{aligned}
$$

Using Bayes Rule

If we have some data $\mathbf{y}$, we can write down a 'likelihood function':

$$p(\mathbf{y}|\theta)$$

This is 'the probability of the data given the parameter'
We can use Bayes Rule to get the pdf of a parameter given the data.

$$p(\theta|\mathbf{y}) = \frac{p(\theta)p(\mathbf{y}|\theta)}{p(\mathbf{y})} = \frac{p(\theta)p(\mathbf{y}|\theta)}{\int_{\theta\in\mathbb{R}} p(\theta)p(\mathbf{y}|\theta)d\theta}$$

Where $p(\theta)$ is the prior.

Inference with a Non-informative Prior

Suppose that we have a little information about something, we might want to put a uniform prior on $\theta = (\mu, \sigma^2)$. This means a uniform constant on μ, and (perhaps) a prior on σ^2 that puts more weight on lower values of σ^2. We can then say that:

$$p(\mu,\ log\sigma^2) \propto 1$$

$$p(\mu,\ \sigma^2) \propto \frac{1}{\sigma^2}$$

Notice that this prior is not a real probability. These are called improper priors.

Posteriors for Normal data with Non-informative Priors

If $y_1, ...y_n$ are iid Normal($\mu,\ \sigma^2$) with the above prior:

a) $\sigma^2|\mathbf{y} \sim$ Scaled Inverse Chi-squared$(n-1, s^2)$
   dinvchisq($\bar{y}$, n-1, s^2)

b) $\mu|\sigma^2,\ \mathbf{y} \sim$ Normal $(\bar{y},\ \sigma^2/n)$

c) $\mu|\mathbf{y} \sim t_{n-1}(\bar{y},\ s^2/n)$
   dt($(y+\bar{y})/sqrt(s^2/n)$, n-1)

Where $s^2 = \sum_{i=1}^{n}(y_i - \bar{y})^2/(n-1)$

Example: Voter Fraud

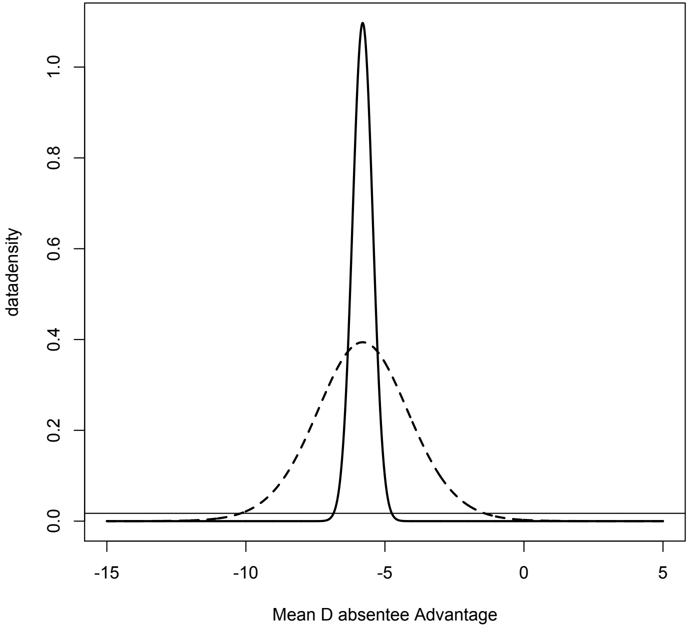
In 1993, Pennsylvania had an election in which the Democratic candidate lost in terms of votes cast in person, but won once absentee ballots were counted.

In a followup study, Ashenfelter studeid 21 elections, and found that on average the gap between the democratic vote share in person and the absentee ballots was -5.8 percentage points and that the sample standard deviation was 55.3

Suppose we use the improper prior, and $\bar{y} = -5.8$, $n = 21$, and $s^2 = (n-1)^{-1} \sum_{i=1}^{n} (y_i - \bar{y})^2 = 58.1$

Using c) from the previous slide:

$$\mu | \mathbf{y} \sim t_{20}(-5.8,\ 58.1/21)$$



datadensity
0.0
0.2
0.4
0.6
0.8
1.0
-15
-10
-5
0
5
Mean D absentee Advantage